Jona Nakai
DS 4420
Homework #3

1.

	I1	I2	I3	I4	I5	I6
User 1	6.6	5	3	10	7	8
User 2	7	4.5	6	4	4.5	1
User 3	5	3	5.8	8	3	10
User 4	6	9	6	5	1	3

	I1	I2	I3	I4	I5	I6
User 1	0	-1.6	-3.6	3.4	0.4	1.4
User 2	2.5	0	1.5	-0.5	0	-3.5
User 3	-0.8	-2.8	0	2.2	-2.8	4.2
User 4	1	4	1	0	-4	-2

	U1	U2	U3	U4
U1	1	-0.48	0.50	-0.43
U2	-0.48	1	-0.62	0.39
U3	0.50	-0.62	1	
U4	-0.43	0.39		1

$$S_c(U1, U2) = \frac{\langle 0, -1.6, -3.6, 3.4, 0.4, 1.4\rangle \cdot \langle 2.5, 0, 1.5, -0.5, 0, -3.5\rangle}{|\langle 0, -1.6, -3.6, 3.4, 0.4, 1.4\rangle| \, |\langle 2.5, 0, 1.5, -0.5, 0, -3.5\rangle|} = -0.4846$$

$S_c(U1, U3) = 0.4967$

$S_c(U1, U4) = -0.4323$

$S_c(U2, U3) = -0.6236$

$S_c(U2, U4) = 0.3894$

U1 min max: min=-0.4846 max=0.4967

	U2	U3	U4
U1	0	1	0.0533

U2 min max: min=-0.6236 max:0.3894

	U1	U2	U3
U2	0.1372	0	1

$$r(U1, I1) = \frac{1 \cdot 5 + 0.0533 \cdot 6}{1 + 0.0533} = 5.05$$

$$r(U2, I2) = \frac{1 \cdot 9 + 0.1372 \cdot 5}{1 + 0.1372} = 8.52$$

$$r(U3, I3) = 3.77$$

$$r(U4, I4) = 4.76$$

$$r(U2, I5) = 1.72$$

3a)

	I1	I2	I3	I4	I5	I6	
User 1	6	5	3	10	7	8	
User 2	7	5.67	6	4	3.67	1	
User 3	5	3	5	8	3	10	
User 4	6	9	6	7.33	1	3	← 5.5

	I1	I2	I3	I4	I5	I6
User 1	0	-0.67	-2	2.67	3.33	2.5
User 2	1	0	1	-3.33	0	-4.5
User 3	-1	-2.67	0	0.67	-0.67	4.5
User 4	0	3.33	1	0	-2.67	-2.5

$$S_c(I_1, I_2) = \frac{\langle 0,1,-1,0\rangle \cdot \langle -0.67, 0, -2.67, 3.33\rangle}{|\langle 0,1,-1,0\rangle| \, |\langle -0.67, 0, -2.67, 3.33\rangle|} = 0.6247$$

	I1	I2	I3	I4	I5	I6
I1	1	.62				
I2	.62	1				
I3	.70	.45				
I4	-.83	.06				
I5	.24	-.74				
I6	-.93	-.26				

	I_2	I_3	I_4	I_5	I_6
I1	0.95	1	0.06	0.71	0

	I_1	I_3	I_4	I_5	I_6
I_2	0.62	0.45	0.06	-0.74	-0.26

$$r(U1, I1) = \frac{1 \cdot 3 + 0.95 \cdot 5}{1 + 0.95} = 3.97$$

$$r(U2, I2) = \frac{1 \cdot 7 + 0.87 \cdot 6}{1 + 0.87} = 6.53$$

$r(U3, I3) = 4.03$

$r(U4, I4) = 1.98$

$r(U2, I5) = 2.69$